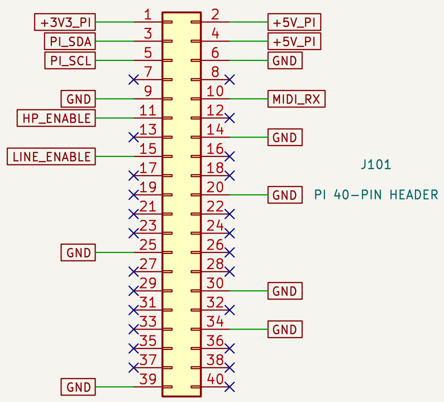
ARDOR / CODEC ZERO I/O

01 / System interface and power

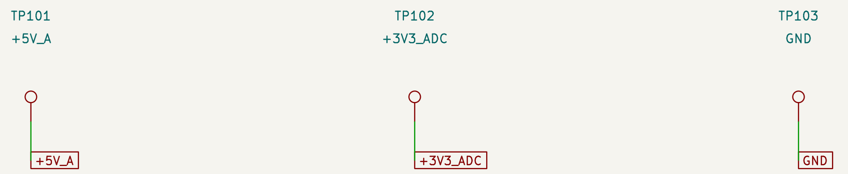
REV A | ENGINEERING PROTOTYPE — VALIDATE BEFORE PRODUCTION

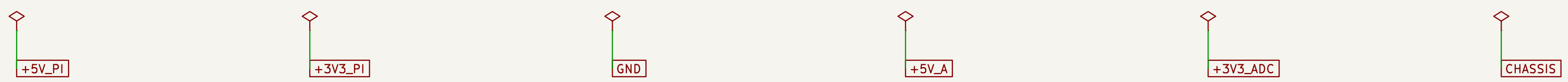
PI HEADER / STACKING INTERFACE



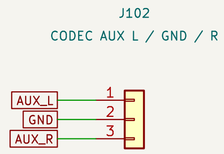
GPIO15 / pin 10: UART RX, 31250 baud, 8-N-1
GPIO17 / pin 11: headphone enable, active high
GPIO27 / pin 13: reserved for Codec Zero button

GPIO22 / pin 15: line relay enable, active high
I2C1: GPIO2/3. ADS1115 address 0x48.
Reserved Codec pins: GPIO18-21 I2S; GPIO23/24 LEDs;
GPIO27 button; GPIO0/1 HAT EEPROM. No added loads.



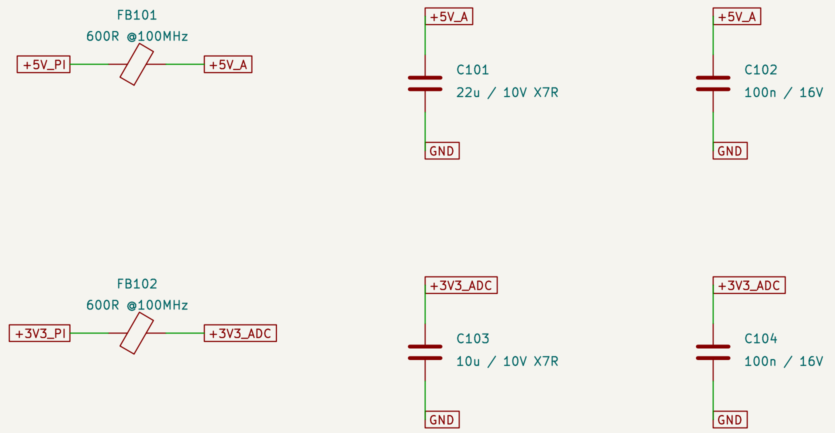


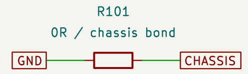
AUX HEADER HARNESS



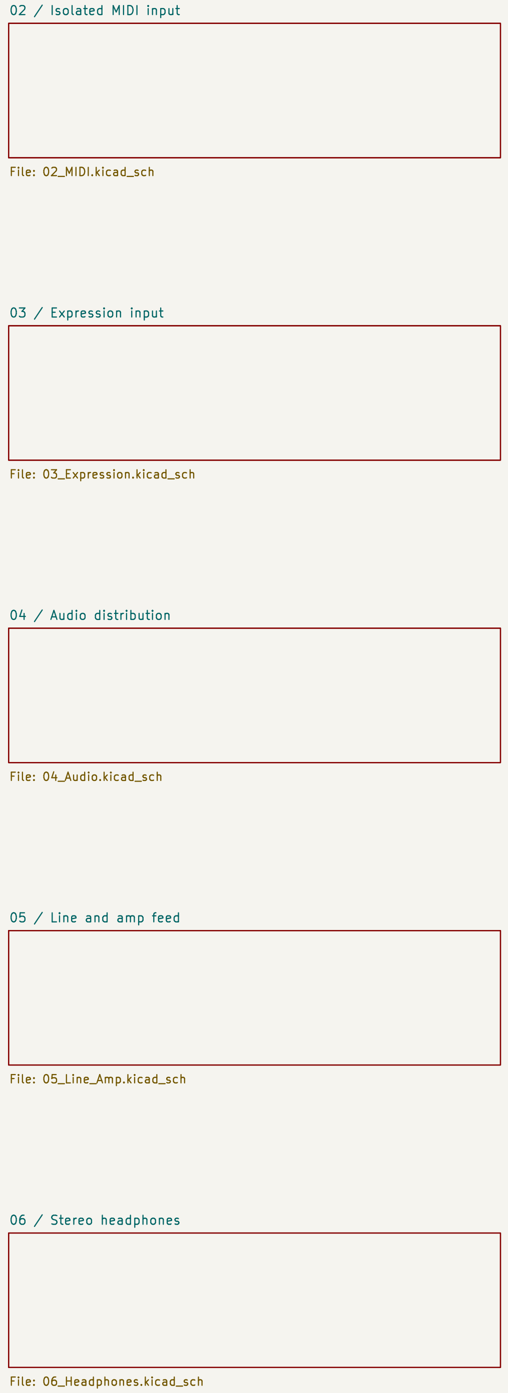
J102 pin order is OUR harness definition.
Map to the actual Codec Zero L/R/GND pads.
Never connect the mono speaker BTL output here.

FILTERED BRANCHES / 5 V REGULATED INPUT ONLY





Bond enclosure at connector entry. Short, wide copper.
Use one continuous PCB ground plane; place by function.
Do not route ESD or charge-pump return through ADC/audio.
5 V source: 4.75–5.25 V; reserve 150 mA beyond Pi/Codec.
Pi provides 3.3 V. No second supply or USB back-feed.



Prototype design; see DESIGN_NOTES.md for release checks
ARDOR — Codec Zero expansion
Sheet: /
File: Ardor_IO.kicad_sch

Title: 01 / System interface and power

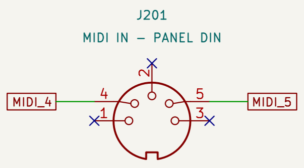
Size: A3	Date: 2026-09-05	Rev: A
KiCad E.D.A. 9.0.2		Id: 1/6

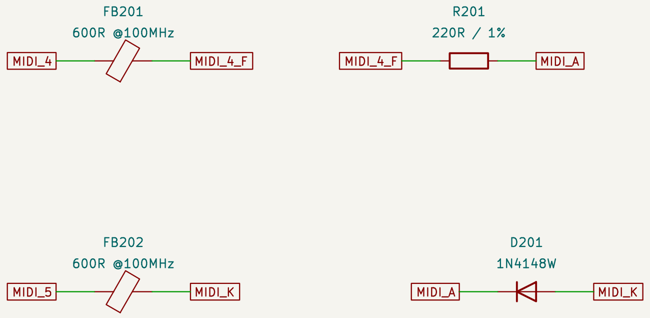
ARDOR / CODEC ZERO I/O

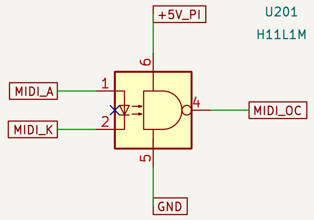
02 / Isolated MIDI input

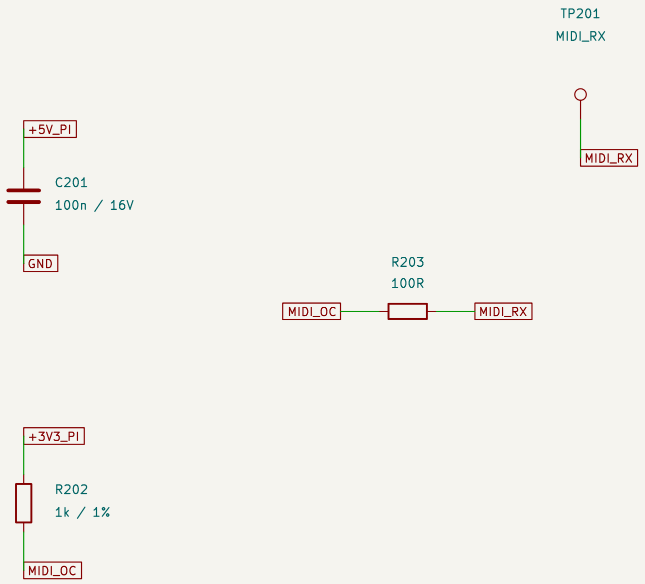
REV A | ENGINEERING PROTOTYPE — VALIDATE BEFORE PRODUCTION

5–PIN DIN / FLOATING CURRENT LOOP

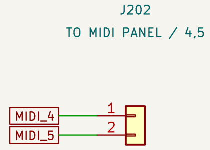






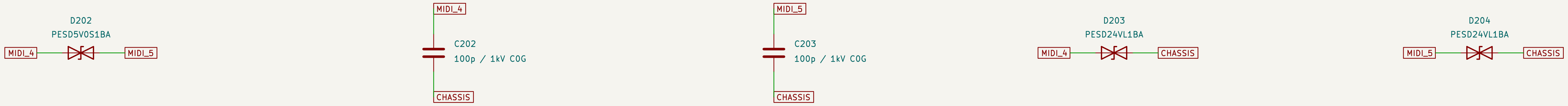


Female DIN, 180 degrees. Use contact numbers,
not an assumed solder–side drawing.
Pins 1, 2, 3 have NO PCB connection.
Metal shell bonded to enclosure mechanically.



MIDI LED on –> UART low. No inverter required.
Open collector pulled up ONLY to 3.3 V.
Disable serial console; enable stable UART RX.
Never use a 5 V pull–up on Raspberry Pi GPIO.

CONNECTOR ESD / RF NETWORK



D202 clamps differential transients; D203/D204 divert common–mode ESD to CHASSIS (24 V standoff).
Keep the MIDI loop copper isolated from logic GND: >= 3 mm clearance around U201 input and nets.
Common–mode range is limited by D203/D204. Functional ground–loop isolation; NOT safety isolation.
No MIDI–pin connection to logic supply. Place TVS and 100p RF capacitors at DIN harness entry.
Qualify IEC 61000–4–2 on the assembled enclosure, including common–mode strikes and UART recovery.

Prototype design; see DESIGN_NOTES.md for release checks
ARDOR — Codec Zero expansion
Sheet: /02 / Isolated MIDI input/
File: 02_MIDI.kicad_sch

Title: 02 / Isolated MIDI input

Size: A3	Date: 2026–09–05	Rev: A
KiCad E.D.A. 9.0.2		Id: 2/6

ARDOR / CODEC ZERO I/O

03 / Expression input

REV A | ENGINEERING PROTOTYPE – VALIDATE BEFORE PRODUCTION

PASSIVE TRS PEDAL / TIP–RING SELECT

J301
EXPRESSION / 6.3mm TRS
S GND
R EXP_RING
T EXP_TIP

JP301
POLARITY / 2 SHUNTS
EXP_TIP 1 2 EXP_RING
EXP_WIPER 3 4 EXP_EXC
EXP_RING 5 6 EXP_TIP

R301
1k / 1%
+3V3_ADC EXP_EXC

R302
10k / 1%
EXP_WIPER EXP_ADC

EXP_WIPER
R303
1M
GND

EXP_ADC
C301
100n / 16V
GND

D303
BAT54H
+3V3_ADC EXP_ADC

D304
BAT54H
EXP_ADC GND

TP301
EXP_ADC
EXP_ADC

Normal: shunts 1–3 and 2–4 (tip=wiper).
Reverse: shunts 3–5 and 4–6 (ring=wiper).
Fit BOTH shunts in the same position.
Sleeve = ground. 10k–100k linear pot.

J302
TO EXP PANEL / T,R,S
EXP_TIP 1
EXP_RING 2
GND 3

D301
PESD5V0S1BA
EXP_TIP CHASSIS

D302
PESD5V0S1BA
EXP_RING CHASSIS

+3V3_ADC
U301
ADS1115IDGS
8 VDD
2 ALERT/RDY
EXP_ADC 4 AIN0
EXP_REF_ADC 5 AIN1
GND 6 AIN2
GND 7 AIN3
SCL 10 ADC_SCL
SDA 9 ADC_SDA
ADDR 1 GND
3 GND
GND

R305
33R
ADC_SCL PI_SCL

R306
33R
ADC_SDA PI_SDA

TVS at jack entry; R302 then clamp pair at ADC.
R301 limits insertion/short current to <=3.5 mA.
Passive pedals only; no powered CV source.
Calibrate endpoints to remove pot loading error.

D305
BAT54H
+3V3_ADC EXP_REF_ADC

D306
BAT54H
EXP_REF_ADC GND

R304
10k / 1%
EXP_EXC EXP_REF_ADC

EXP_REF_ADC
C302
100n / 16V
GND

+3V3_ADC
C303
100n / 16V
GND

+3V3_ADC
R307
2.2k / rail bleed
GND

I2C address 0x48; internal reference; PGA +/–4.096 V; 128 SPS. Read AIN0 and AIN1 single–ended.
Use AIN0 / AIN1 ratio, endpoint calibration, smoothing and deadband. No added I2C pull–ups (Pi has them).

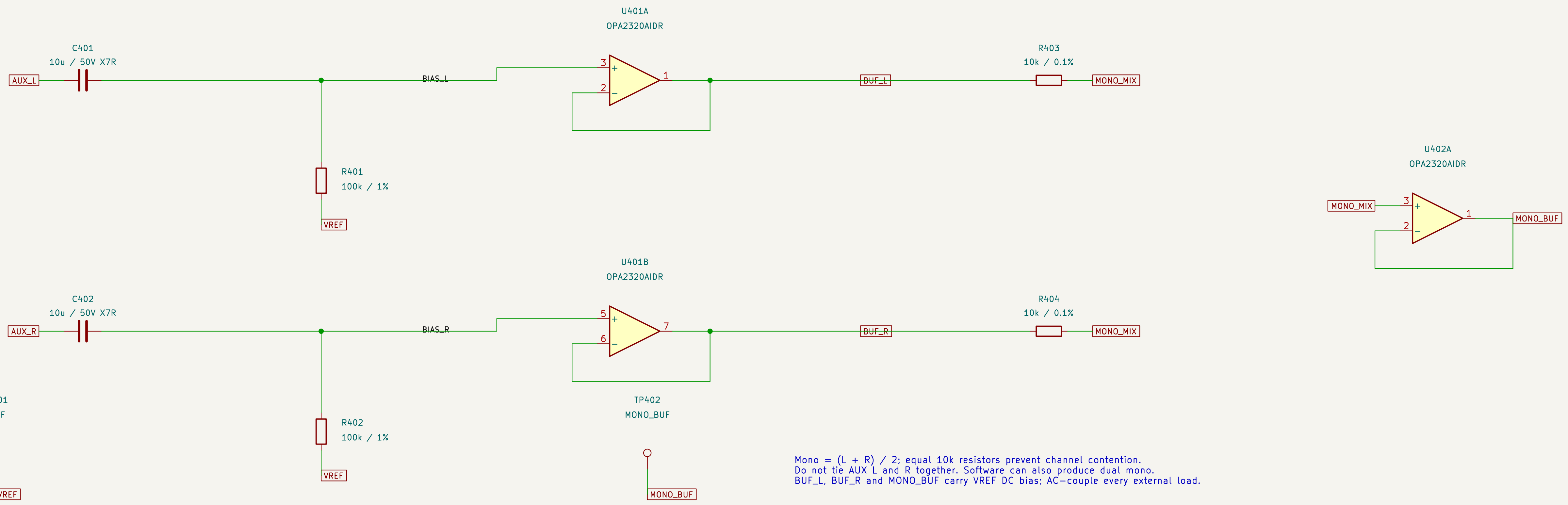
Prototype design; see DESIGN_NOTES.md for release checks
ARDOR – Codec Zero expansion
Sheet: /03 / Expression input/
File: 03_Expression.kicad_sch
Title: 03 / Expression input
Size: A3 | Date: 2026-09-05 | **Rev: A**
KiCad E.D.A. 9.0.2 | Id: 3/6

ARDOR / CODEC ZERO I/O

04 / Audio distribution

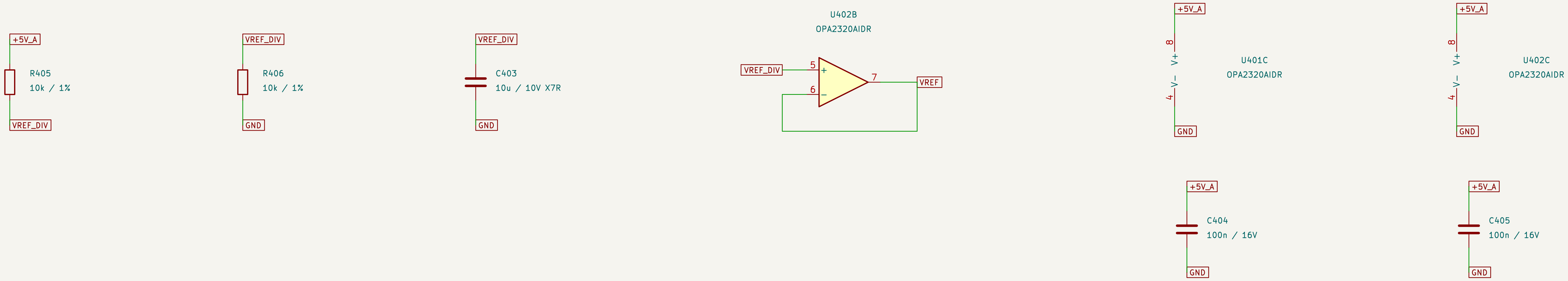
REV A | ENGINEERING PROTOTYPE — VALIDATE BEFORE PRODUCTION

AC—COUPLED STEREO INPUT / UNITY BUFFERS



Mono = (L + R) / 2: equal 10k resistors prevent channel contention.
Do not tie AUX L and R together. Software can also produce dual mono.
BUF_L, BUF_R and MONO_BUF carry VREF DC bias; AC—couple every external load.

MIDRAIL / LOCAL BYPASS



Nominal audio target <=1.0 Vrms per AUX channel. Unity gain throughout.
OPA2320: RRIO on 5 V, unity stable; 2.5 V bias permits bipolar audio swing.
2.2u/100k input pole ~0.72 Hz. Keep VREF local; no large capacitor directly on buffer output.

Prototype design; see DESIGN_NOTES.md for release checks
ARDOR — Codec Zero expansion
Sheet: /04 / Audio distribution/
File: 04_Audio.kicad_sch

Title: 04 / Audio distribution

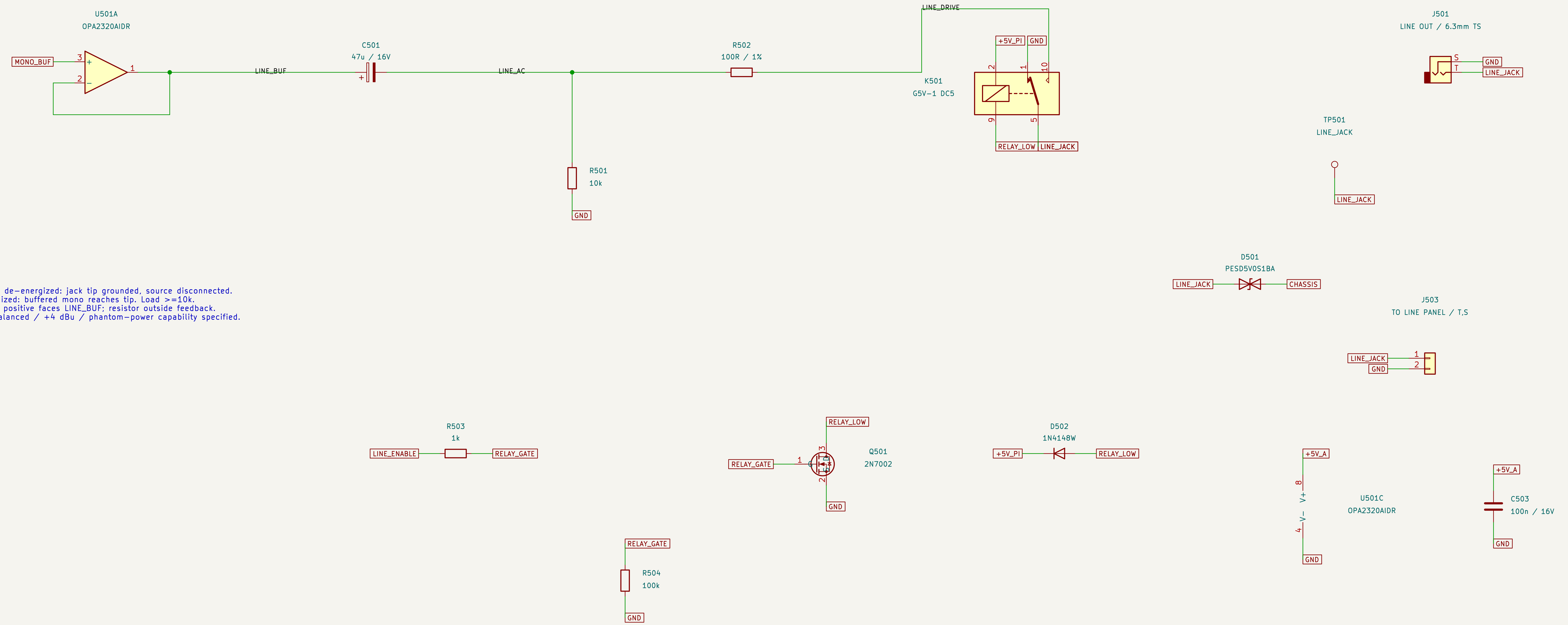
Size: A3	Date: 2026-09-05	Rev: A
KiCad E.D.A. 9.0.2		Id: 4/6

ARDOR / CODEC ZERO I/O

05 / Line and amp feed

REV A | ENGINEERING PROTOTYPE — VALIDATE BEFORE PRODUCTION

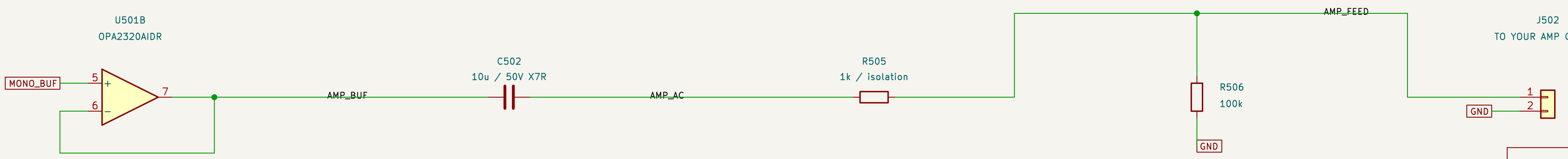
MONO LINE / DC BLOCK / STARTUP RELAY MUTE



Relay de-energized: jack tip grounded, source disconnected.
Energized: buffered mono reaches tip. Load >=10k.
C501 positive faces LINE_BUF; resistor outside feedback.
No balanced / +4 dBu / phantom-power capability specified.

Amp input >=100k. Add your gain, output connector, external-port protection and mute in that project.
J502 is internal, NOT a speaker / amp output jack.

AMP CIRCUIT PLACEHOLDER / INTERNAL HARNESS



Prototype design; see DESIGN_NOTES.md for release checks
ARDOR – Codec Zero expansion
Sheet: /05 / Line and amp feed/
File: 05_Line_Amp.kicad_sch

Title: 05 / Line and amp feed

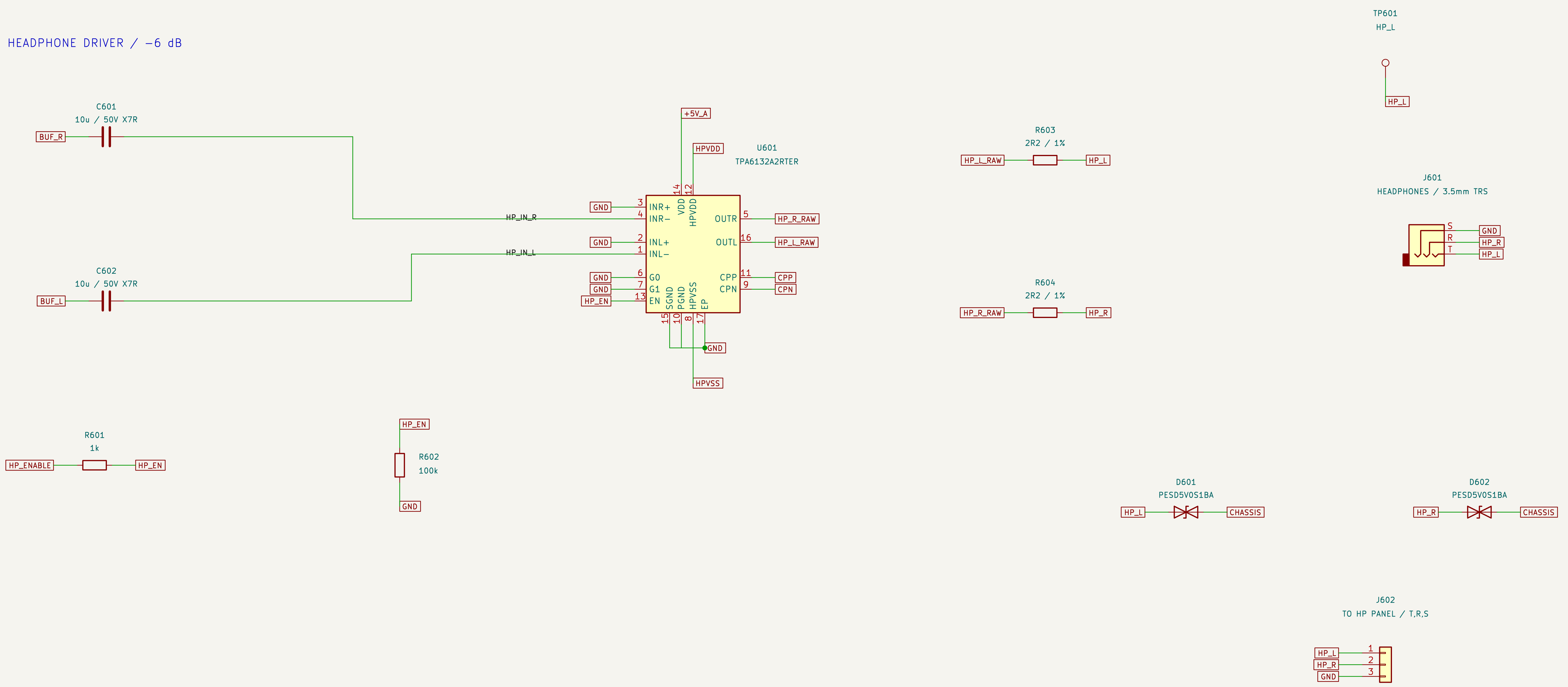
Size: A3	Date: 2026-09-05	Rev: A
KiCad E.D.A. 9.0.2		Id: 5/6

ARDOR / CODEC ZERO I/O

06 / Stereo headphones

REV A | ENGINEERING PROTOTYPE — VALIDATE BEFORE PRODUCTION

STEREO DIRECTPATH HEADPHONE DRIVER / −6 dB



CHARGE PUMP / BYPASS — FOLLOW TI LAYOUT

HPVDD is an INTERNAL regulator output. Never connect it to +5V_A. HPVSS is negative; use nonpolar ceramic.
G0=0, G1=0 gives −6 dB inverting gain. INL+ / INR+ grounded per TI single-ended application.
32 ohm or higher stereo headphones. 1 Vrms AUX gives ~0.47 Vrms at 32 ohm (~6.8 mW); start volume low.
Keep EN low until codec routing is stable; ramp volume. Short, separate SGND jack return to local ground plane.
Place pump capacitors at pins; keep switching loops away from AUX, VREF and expression input traces.

Prototype design; see DESIGN_NOTES.md for release checks
ARDOR — Codec Zero expansion
Sheet: /06 / Stereo headphones/
File: 06_Headphones.kicad_sch
Title: 06 / Stereo headphones
Size: A3 | Date: 2026-09-05 | Rev: A
KiCad E.D.A. 9.0.2 | Id: 6/6